unpredictable which allows us to obtain *new* information. When we acquire new information about a system, the total relevant information cannot grow indefinitely (because of the first postulate), and part of the previous information becomes *irrelevant*, that is to say, it no longer has any effect upon predictions of the future. In quantum mechanics when we interact with a system, we don't only learn something, we also 'cancel' a part of the relevant information about the system.[fn52]

The entire formal structure of quantum mechanics follows in large measure from these two simple postulates. Therefore, the theory lends itself in a surprising way to being expressed in terms of information.

The first to realize that the notion of information was fundamental to the understanding of quantum reality was John Wheeler, the father of quantum gravity. Wheeler coined the phrase 'It from bit' to express this idea, meaning that 'everything is information'.

Information reappears, then, in the context of quantum gravity. Remember: the area of any surface is determined by the spins of the loop which intersect this surface. These spins are discrete quantities, and each one contributes to the area.

A surface with a fixed area may be formed from these elementary quanta of area in many different ways, say, in a number of ways N. If you know the area of a surface but don't know exactly how its quanta of area are distributed, you have missing information about the surface. This is one of the ways of computing the heat of black holes: the quanta of area of a black hole enclosed in a surface of a certain area can be in N different possible distributions. It is like for the cup of tea, in which the molecules can move in N different possible ways. Thus we can associate a quantity of missing information, that is to say, entropy, with a black hole.

The amount of information associated thus with a black hole depends directly upon the area A of the hole: the larger the hole, the greater the amount of missing information.

When information enters into a black hole, it is no longer recoverable from outside. But the information which enters the black hole carries with it the energy by which the black hole becomes larger and increases its area. Viewed from outside, the information lost in the black hole now appears as entropy associated with the area of the hole. The first to suspect something similar was the Israeli physicist Jacob Bekenstein.

But the situation is anything but clear because, as we have seen in the last chapter, black holes emit thermal radiation and very slowly evaporate, becoming smaller and smaller until they probably disappear, subsumed in that ocean of microscopic black holes which constitutes space at the Planck scale. Where does the information that has fallen into the black hole as the black